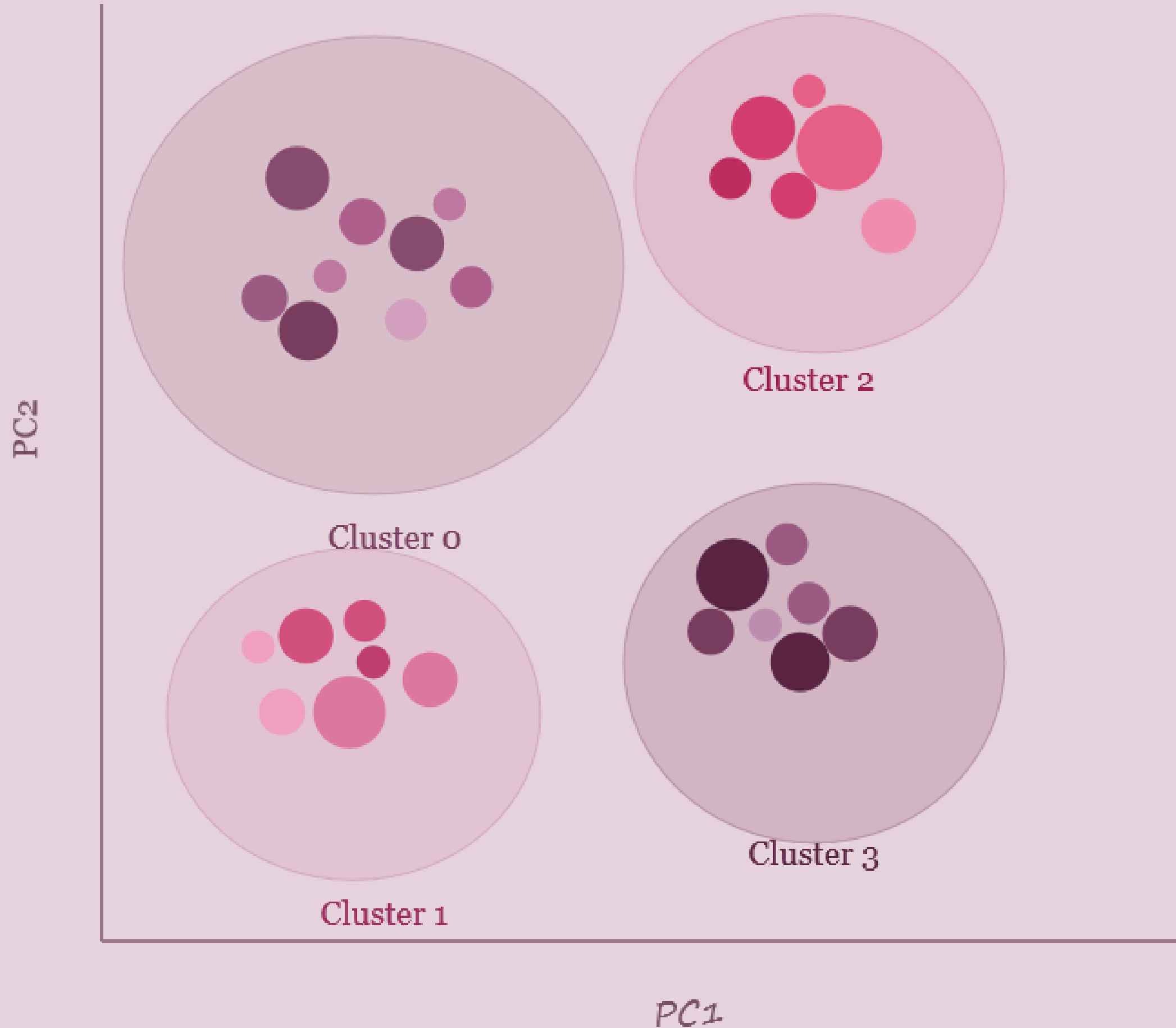




Client Segmentation

PRESENTED BY

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K-Means · PCA · Isolation Forest

Table of Contents

1

Dataset Overview

2

Exploratory Data Analysis

Univariate • Bivariate • Domain-Specific

3

Data Cleaning

Duplicates • Missing Values • Anomalies

4

Feature Engineering

RFM Construction • Behavioral Features

5

Outlier Detection & Treatment

Isolation Forest • Post-Cleaning Correlation

6

Feature Scaling

7

Dimensionality Reduction

Principal Component Analysis (PCA)

8

K-Means Clustering

Elbow • Silhouette • Cluster Profiles

9

Recommendation System

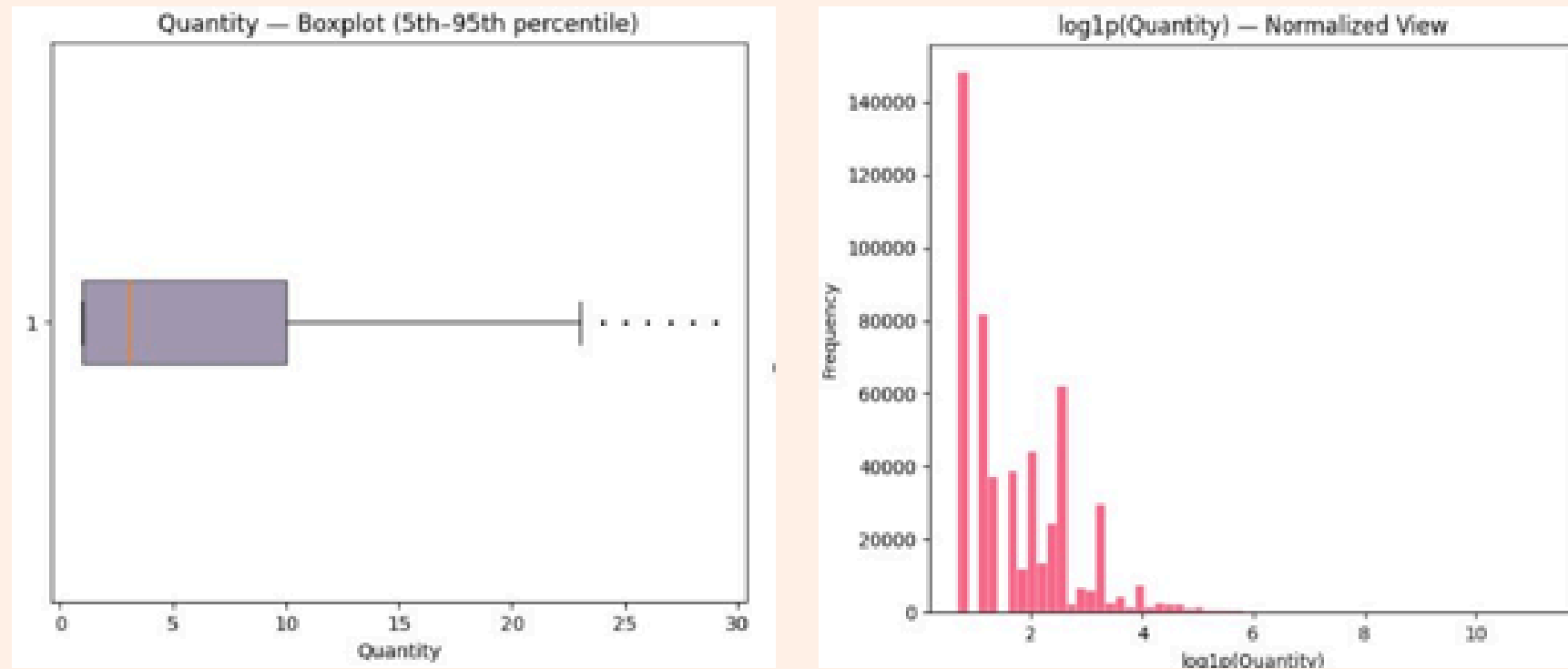
1. Dataset Overview

Variable	Description
InvoiceNo	Code representing each unique transaction. If this code starts with letter 'c', it indicates a cancellation.
StockCode	Code uniquely assigned to each distinct product.
Description	Description of each product.
Quantity	The number of units of a product in a transaction.
InvoiceDate	The date and time of the transaction.
UnitPrice	The unit price of the product in sterling.
CustomerID	Identifier uniquely assigned to each customer.
Country	The country of the customer.

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	2010-12-01 08:26:00	2.55	17850	United Kingdom
1	536365	71053	WHITE METAL LANTERN	6	2010-12-01 08:26:00	3.39	17850	United Kingdom
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	2010-12-01 08:26:00	2.75	17850	United Kingdom
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	2010-12-01 08:26:00	3.39	17850	United Kingdom
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	2010-12-01 08:26:00	3.39	17850	United Kingdom

Distribution of Numerical Variables :

Quantity Distribution

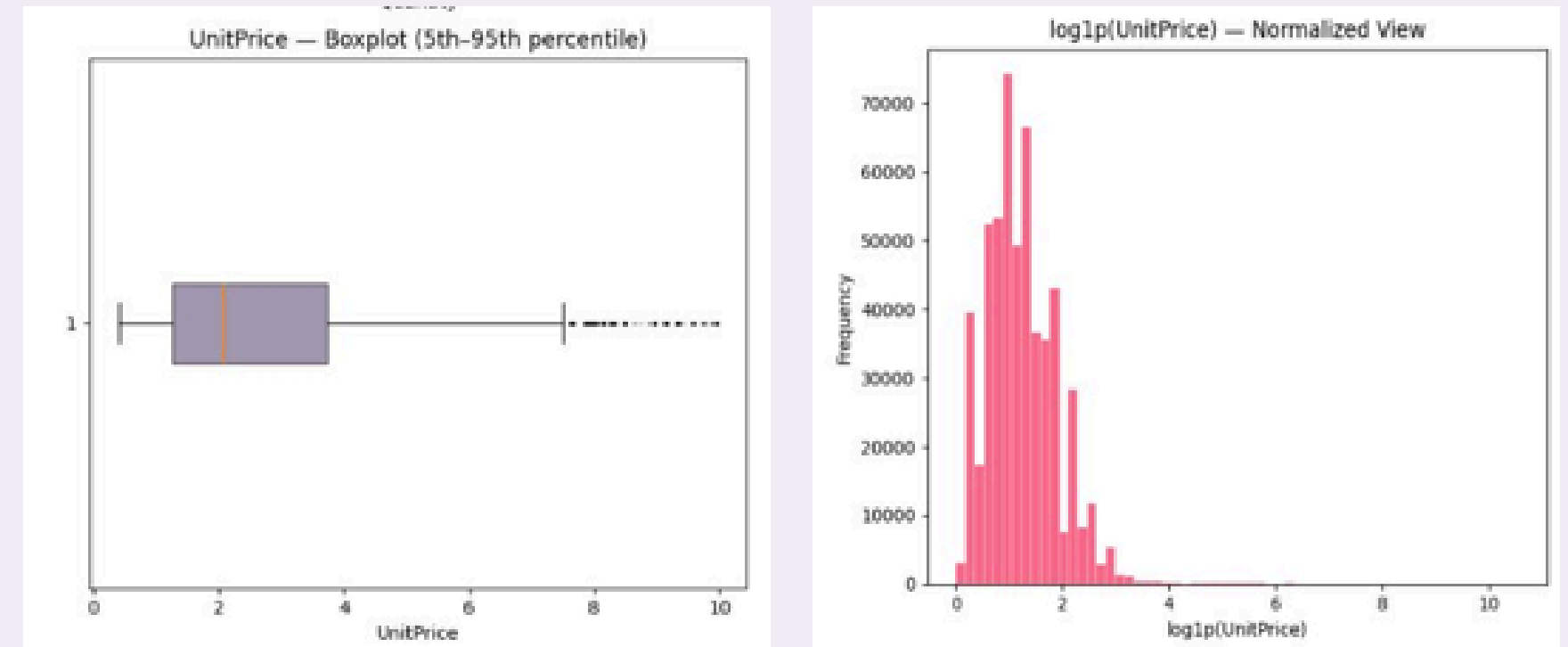


Interpretation & Segment Implication

- Most purchases are small quantities
- Extreme outliers (bulk orders & returns) distort raw distribution
- 10,624 negative values (returns)
- high-quantity outliers likely represent B2B or wholesale customers

Decision → Filter negatives & zeros

UnitPrice Distribution



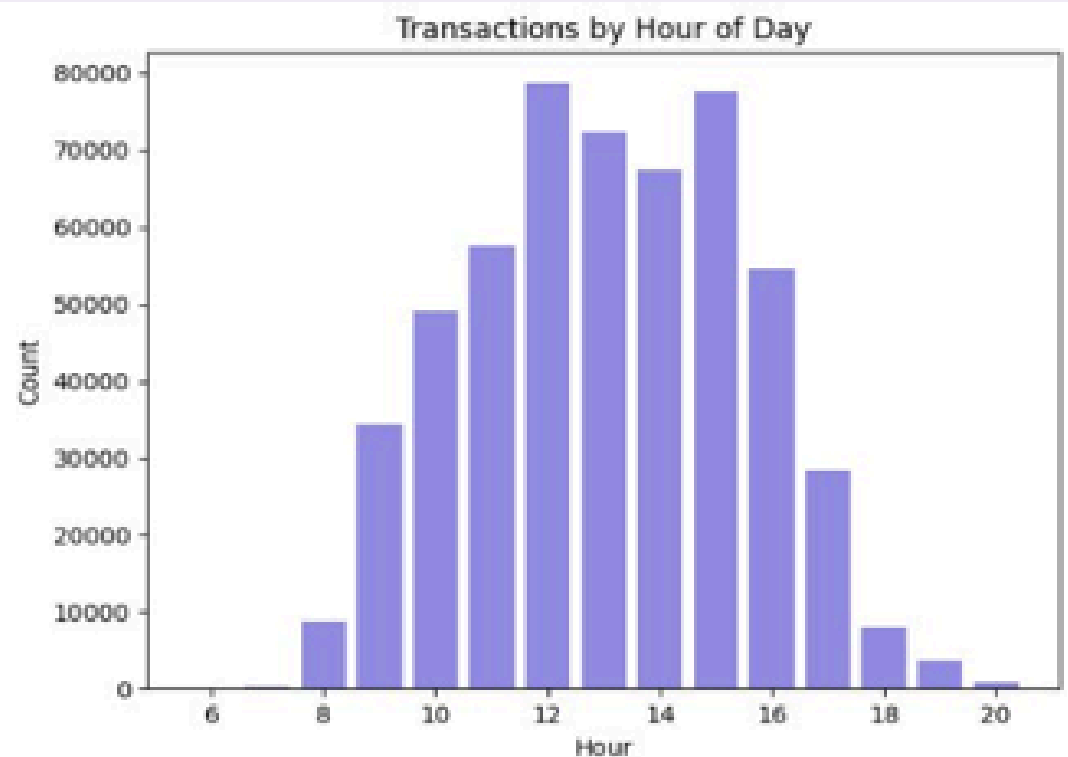
Interpretation & Segment Implication

- Nearly all prices cluster between £0 and £5 — raw mean (£4.61) is pulled far above median (£2.08) by extreme outliers.
- 2 negative values and 2,515 zero prices, they do not represent real product transactions.
- high-price outliers likely signal luxury/premium items, may define a distinct customer segment

Decision → Remove negatives & zero prices

Distribution of Categorical Variables :

🕒 Hour of Day

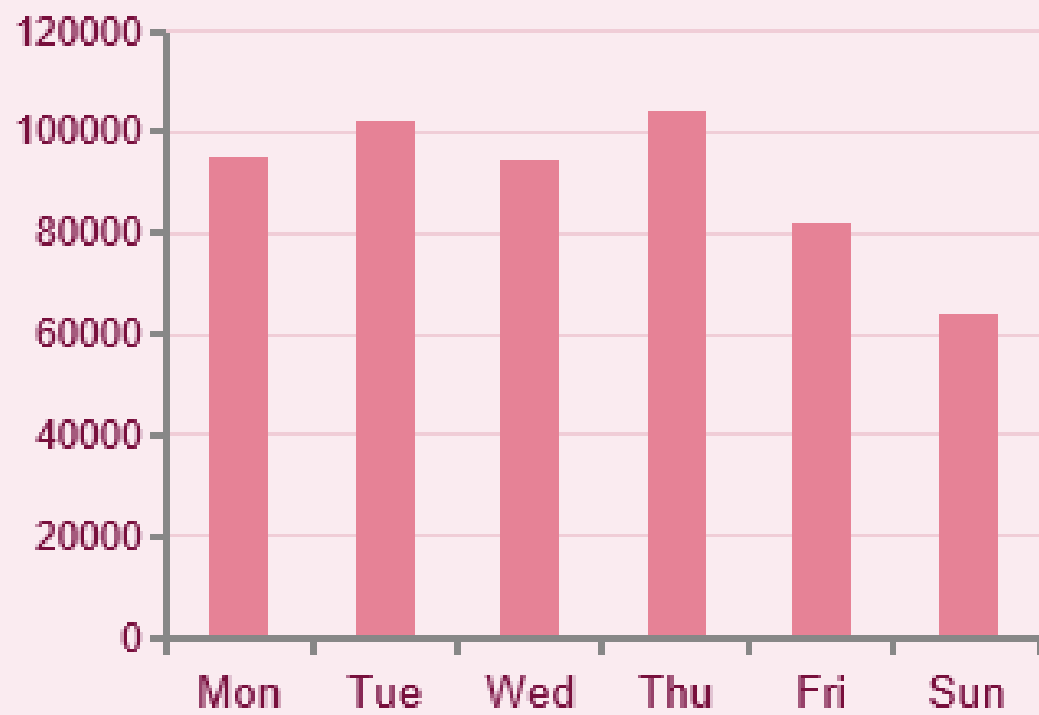


Interpretation & Segment Signal

- Peak 12pm–3pm — customers shop like professionals
- Early-morning shoppers → likely B2B bulk buyers

Feature → Shopping_Hour

📅 Day of Week

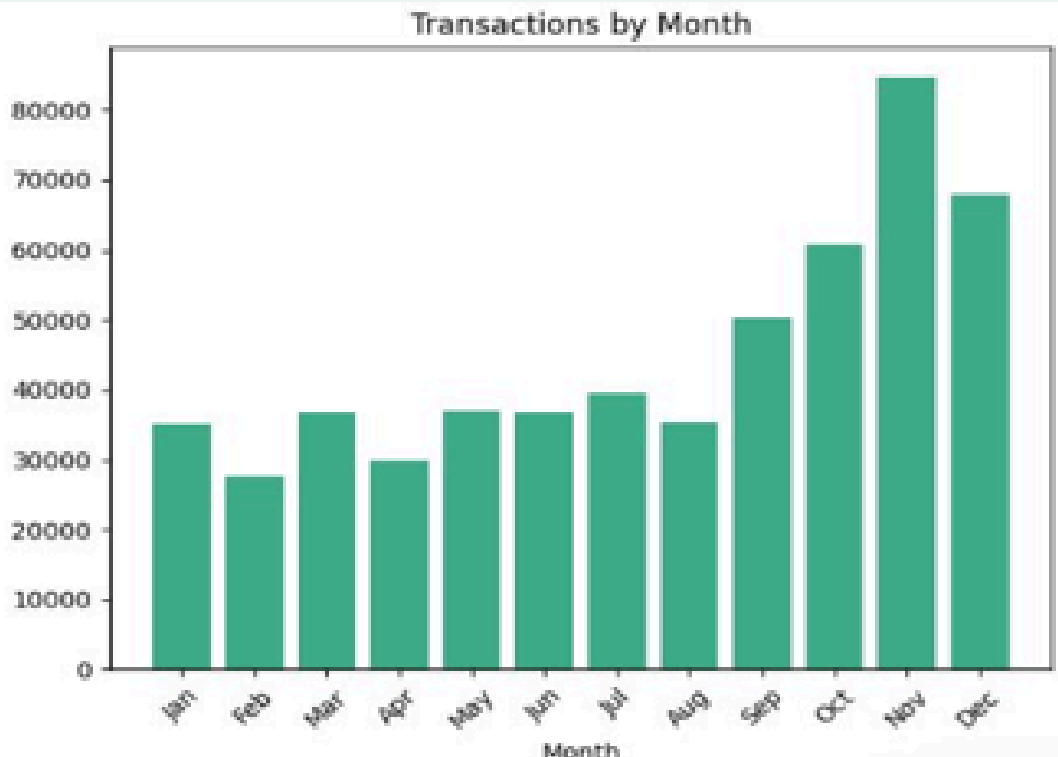


Interpretation & Segment Signal

- Mon–Thu dominates — business-driven rhythm
- Favourite_Shopping_Day separates habitual from spontaneous buyers

Feature → Favourite_Shopping_Day

📅 Month



Interpretation & Segment Signal

- Oct–Nov surge → 2 profiles: Seasonal vs Year-Round
- Recency scores skewed by Q4 — model must account for this

Feature → Seasonality_Flag

Bivariate Analysis :

Monthly Revenue Trend (£k)



Peak Revenue **£1.47M**

Flat Period

Dec'10 –
Aug'11

Dec'11 Drop

Data
truncation

Interpretation & Segment Implication

- Revenue flat at £500k–£800k (Dec'10–Aug'11), then sharply accelerates from Sep, peaking at £1.47M in Nov 2011
- Dec 2011 steep drop is a data truncation effect — not a real business decline
- Clear seasonal cycle: Q4-only customers represent a fundamentally different profile from year-round consistent buyers
- Segment implication: purchase timing will be a key differentiator — seasonal vs. steady buyers must be modelled separately

Feature → Seasonality_Flag • Adjust Recency for Q4 seasonal buyers

Spearman Correlation



Spearman $r = -0.39 \rightarrow$ Moderate negative

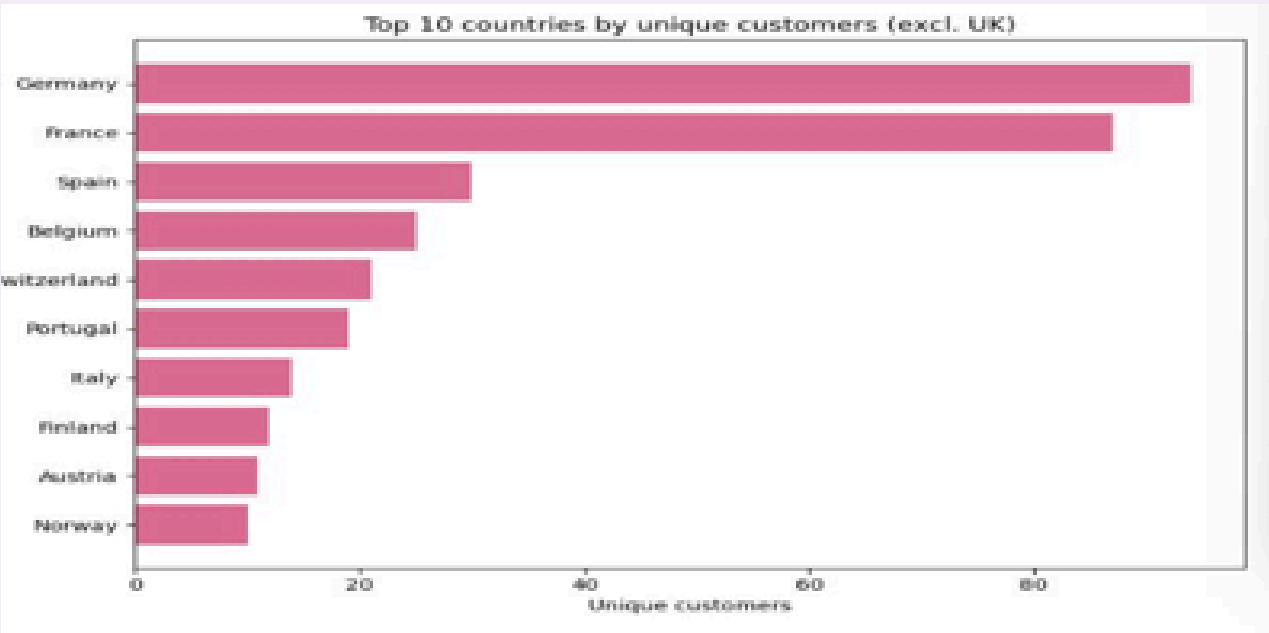
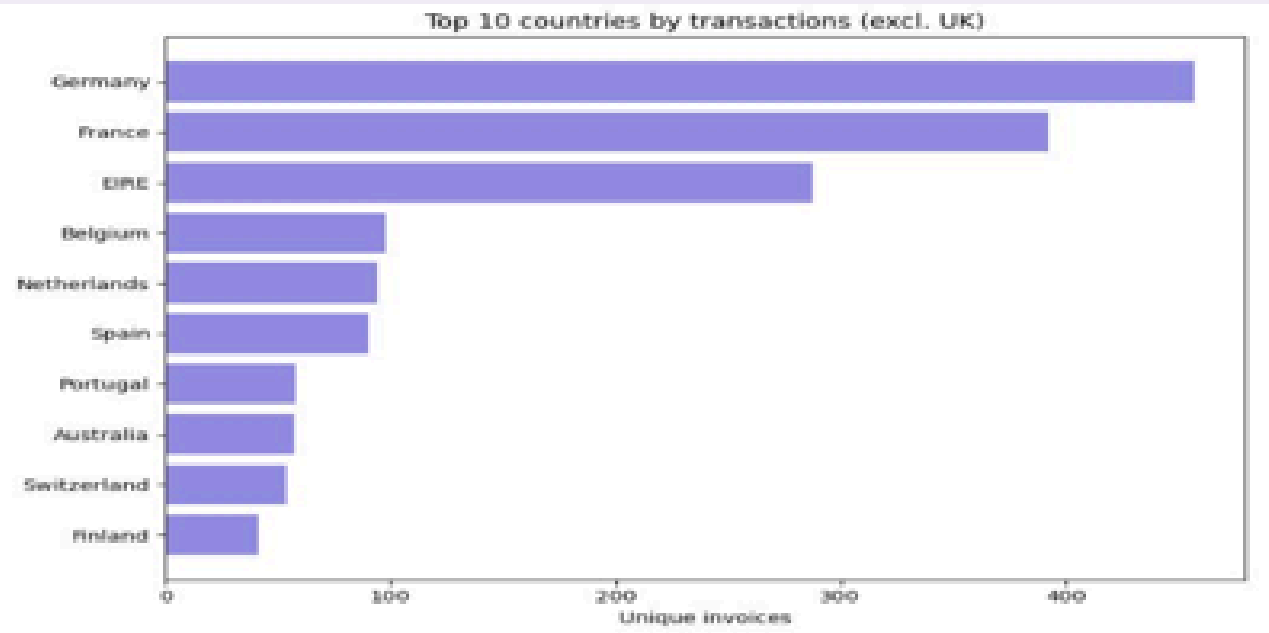
What This Means

- ↑ Quantity = ↓ UnitPrice — bulk buyers pay less per unit; premium buyers buy fewer, costlier items
- Two distinct buying behaviours: volume-driven vs. value-driven — a single metric can't capture both
- Quantity & UnitPrice are complementary — both must be preserved as separate features
- Segment implication: bulk buyers vs. premium buyers will naturally emerge as distinct clusters

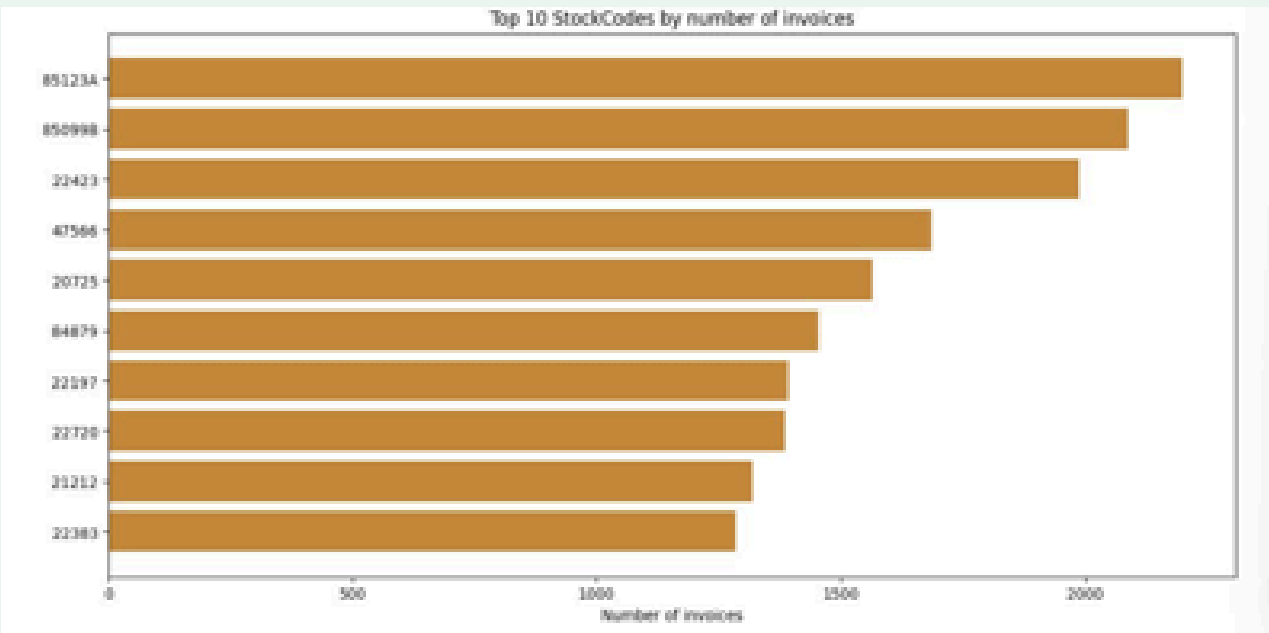
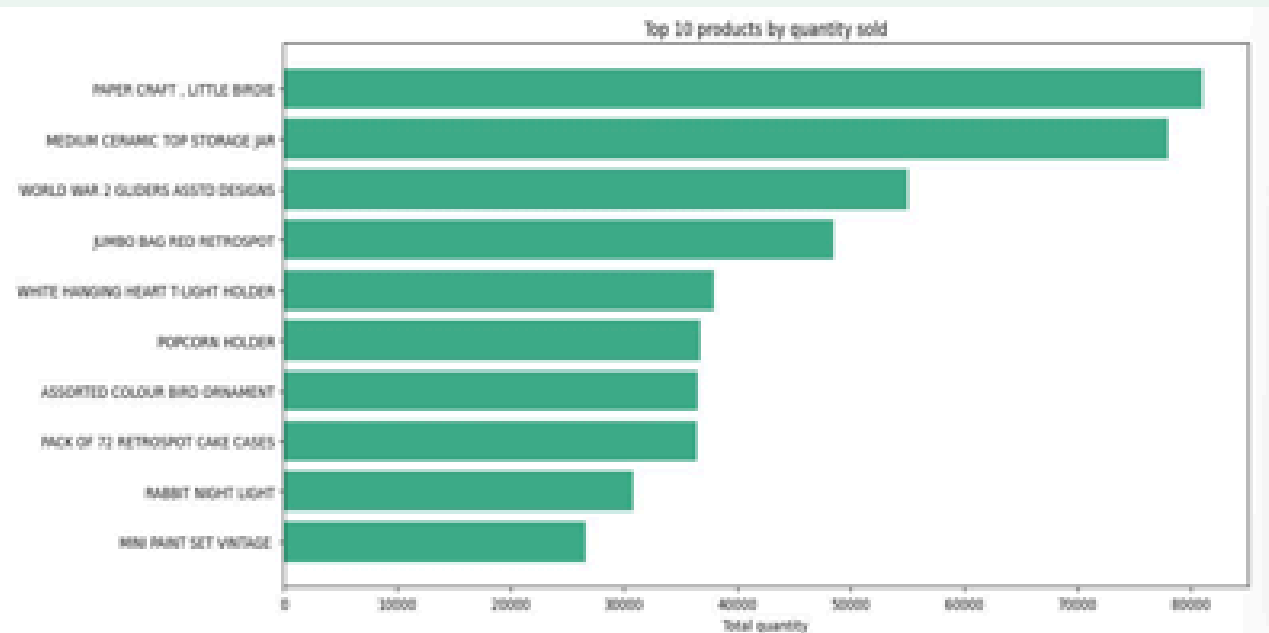
Keep Quantity & UnitPrice as separate cluster features

Geographic and Product Analysis:

Geographic Analysis



Product Analysis



Price Variability

3,366

StockCodes sold
at multiple prices

DOT 686x

M 131x

POST 53x

79321 15x

Same product → different spend per
customer
Interpret monetary features carefully

Cancellations and Zero-Price transactions :

CANCELLATION RATE

1.71%

of all invoices

NEGATIVE QUANTITY ROWS

10,624

slightly above cancelled count

ZERO-PRICE TRANSACTIONS

33

likely samples or write-offs

Returns without cancellation prefix

Some returns are recorded without a formal cancellation invoice prefix “C-” which is an anomaly that should be dealt with.

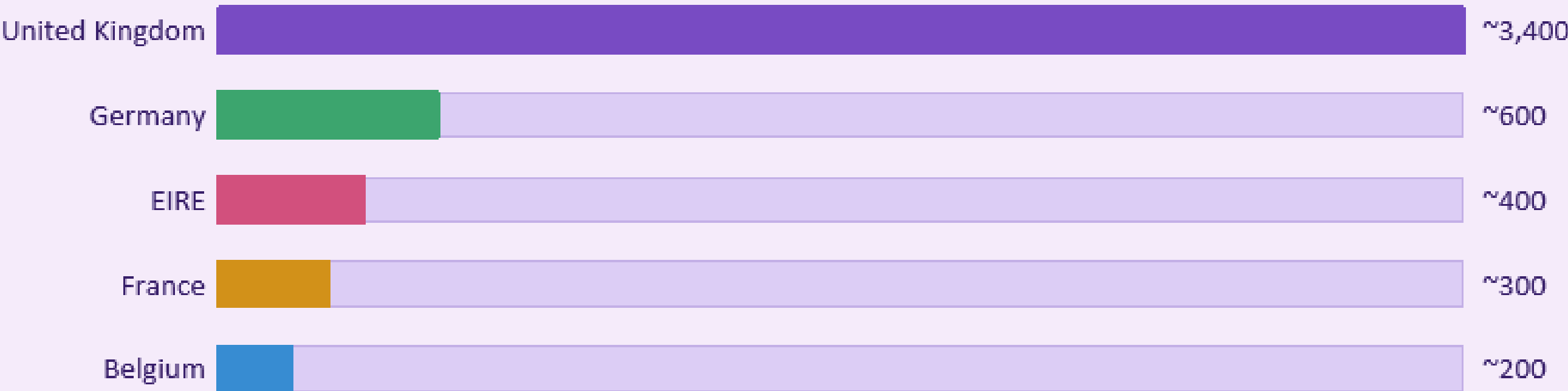
Purchases vs Returns Distribution

Returns are far less frequent than purchases, confirming that returns happen occasionally rather than a systemic issue.

Zero prices: small & isolated

All 33 zero-price rows involve small quantities — likely free samples, internal transfers, or data entry errors.

TOP CANCELLATIONS BY COUNTRY



UK dominates (91.4% of all transactions), so its share of cancellations is expected.

Handling Missing Values

StockCode & Description Inconsistencies:



StockCode & Description Inconsistencies

650

StockCodes with inconsistent descriptions
out of 4,070 unique codes → 16% of products

What this means

- Same StockCode mapped to multiple different descriptions
- Caused by typos, data entry errors, or wrongly marked items
- Makes imputation by StockCode unreliable
- Example: code 20713 has 9 different descriptions (see below)

StockCode: 20713 — 9 different descriptions

→ JUMBO BAG OWLS	→ <code>nan</code>
→ wrongly marked. 23343 in box	→ wrongly coded-23343
→ found / Found	→ wrongly marked 23343
→ Marked as 23343	→ wrongly coded 23343



Missing Values

Only 2 columns out of 8 contain missing values:

Column	Missing	Impact
CustomerID	25.16%	Critical — 1 in 4 transactions has no customer identity
Description	0.27%	Negligible — very few rows affected

CustomerID — 25.16% missing

- Cannot be used in customer segmentation or RFM analysis
- May represent guest checkouts or data entry errors
- Imputing a CustomerID is not possible — no reliable way to assign identity

Decision → Drop all rows where CustomerID is null

Description (0.27%) — Since inconsistent StockCode descriptions make imputation unreliable, rows with missing Description are also dropped.

Handling Duplicates+ StockCode & Description Anomalies:

⚠

StockCode Anomalies

0.48%

of records have anomalous stock codes • 8 out of 3,684

Anomalous codes identified:

→ POST	→ D
→ C2	→ M
→ BANK CHARGES	→ PADS
→ DOT	→ CRUK

- Non-product transactions (postage, bank charges, charity)
- Only 0.48% — but distorts clustering & recommendations
- Introduces noise into customer behaviour analysis

Decision → Exclude anomalous StockCode records

🗒

Handling Duplicates

5,268

duplicate rows detected
1.0% of 541,909 transactions

Duplicate share

99%	1%
-----	----

- Every column identical — cannot be real repeat purchases
- Introduces noise & inaccuracies in clustering

Decision → Remove all identical duplicate rows

➡

Description Column Issues

Entries with lowercase characters — flagged for review:

Flagged entries (lowercase):

→ BAG 125g SWIRLY MARBLES
→ POLYESTER FILLER PAD ...
→ blue and orange
→ Next Day Carriage
→ High Resolution Image

- Non-product names must be removed to avoid polluting analysis
- Service entries (e.g. 'Next Day Carriage') are not real products
- Remaining descriptions standardized to UPPERCASE

Decision → Remove non-product rows • Standardize to UPPERCASE

Feature Engineering

RFM — Classic Features



Recency



Frequency



Monetary

7 Extended Behavioral Features



Customer_Lifetime



Daily_Customer_Value



Average_Basket_Size



Unique_Products_Purchased



Avg_Days_Between_Purchases



Is_UK

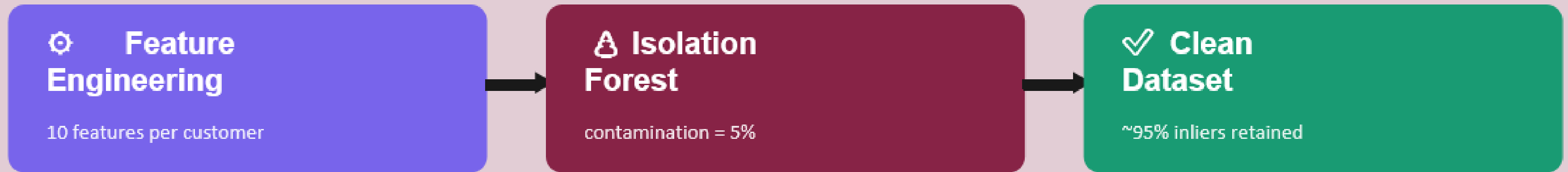


Hour + DayOfWeek

10 features total — a rich behavioural profile far beyond standard RFM, capturing recency, frequency, monetary value, product diversity, timing, geography and lifetime patterns.

Outlier Treatment

Applied after Feature Engineering — using Isolation Forest (contamination = 5%)



Why After Feature Engineering?

Outliers are defined in feature space

Raw columns like Quantity or UnitPrice are poor outlier signals on their own. Anomalies only become visible once behavioural features (Recency, Daily Value, Basket Size...) are computed.

Isolation Forest needs the full profile

The algorithm isolates anomalies by splitting across all 10 features simultaneously. Running it on raw transactions would flag the wrong records.

Protects K-Means centroids

A single extreme B2B customer can pull a centroid away from the true cluster centre. Removing ~5% of anomalous profiles before clustering produces tighter, more meaningful segments.

What Gets Flagged & Impact

~5%

Outliers removed

~95%

Clean inliers kept

Typical flagged profiles:

Bulk Buyers

Extreme bulk buyers — Quantity outliers far above 99th percentile

Big Spenders

Abnormal spenders — Total_Spend 10x median with very few orders

Ghost Customers

Ghost customers — extremely high Recency, zero repeat purchases

Data Anomalies

Data anomalies — unrealistic Daily_Customer_Value ratios

Post-Cleaning Correlation Analysis & Feature Scaling

Results Found

Strongly correlated pairs (Spearman, clean data):

Total_Spend ↔ Total_Purchase		+0.93
Customer_Lifetime ↔ Total_orders		+0.87
Avg_Basket_Size ↔ Avg_Transaction_Value		+0.81
Avg_Days_Between ↔ Daily_Customer_Value		-0.88
Customer_Lifetime ↔ Total_Spend		+0.72
Unique_Products ↔ Total_orders		+0.64

Key finding:

6 pairs with $r > 0.60$ → multicollinearity confirmed → PCA is justified.
Quantity ↔ UnitPrice = -0.39 → complementary, both features kept.

Why Apply StandardScaler?

K-Means is distance-based

K-Means computes Euclidean distances. A feature with large values (Total_Spend in £ thousands) will dominate over one with small values (Total_orders, 1–200) — not because it's more important, but because it's bigger.

PCA is variance-based

PCA finds directions of maximum variance. Without scaling, high-magnitude features capture the first components simply due to their scale — hiding the real structure.

Equal contribution

After scaling all features to $\mu=0$, $\sigma=1$, every feature contributes equally. The model learns from patterns, not magnitudes.

Dimensionality Reduction

Why Apply PCA?

Eliminates multicollinearity

6 feature pairs with $r > 0.60$ create redundant dimensions. PCA rotates the feature space onto orthogonal axes — each component is independent.

Reduces the curse of dimensionality

K-Means degrades in high-dimensional spaces. Fewer, uncorrelated components produce tighter, more meaningful cluster boundaries.

Removes noise dimensions

Low-variance components capture measurement noise rather than real behaviour. Dropping them makes the clustering more robust.

Visualisation

The first 2 principal components (PC1 vs PC2) allow the 4 clusters to be plotted and visually validated in 2D.

6 components

→ 80.90% variance explained

Top component interpretations:

PC1

Volume & Frequency

PC2

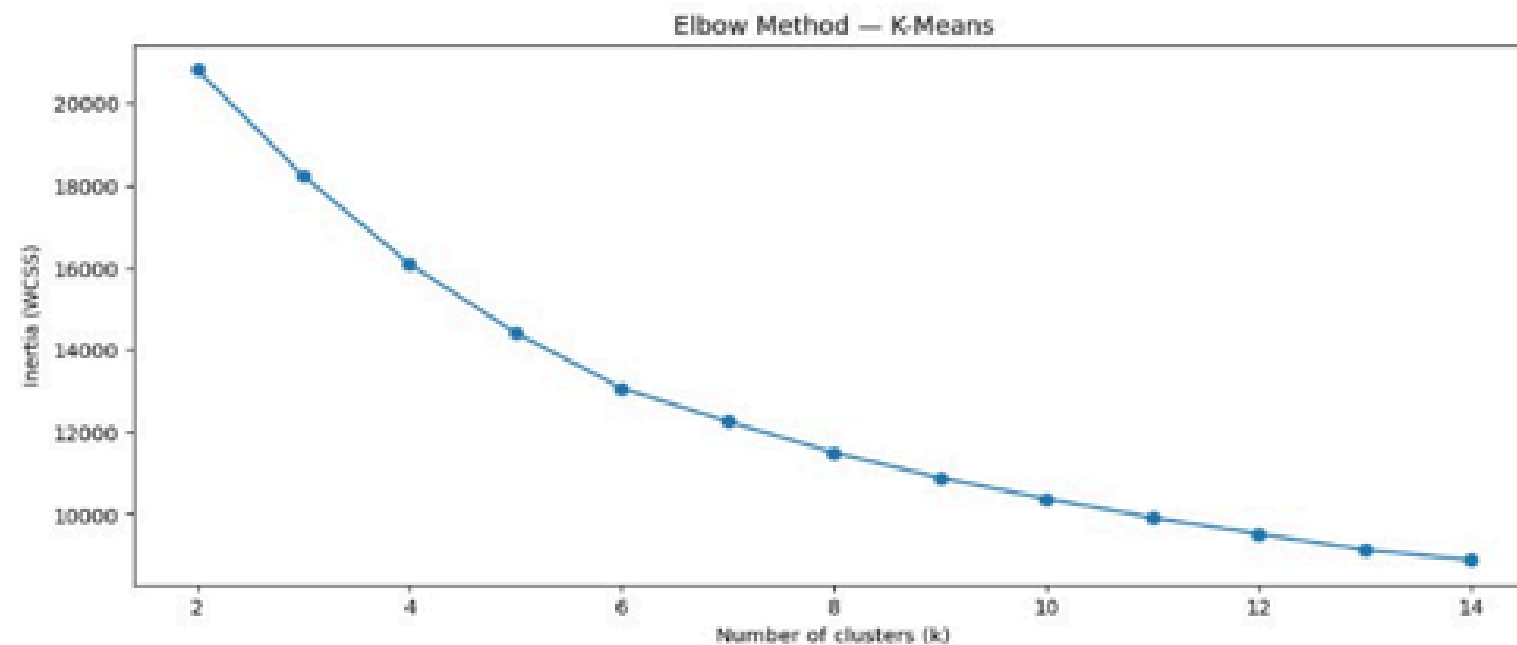
Purchase Timing

PC3

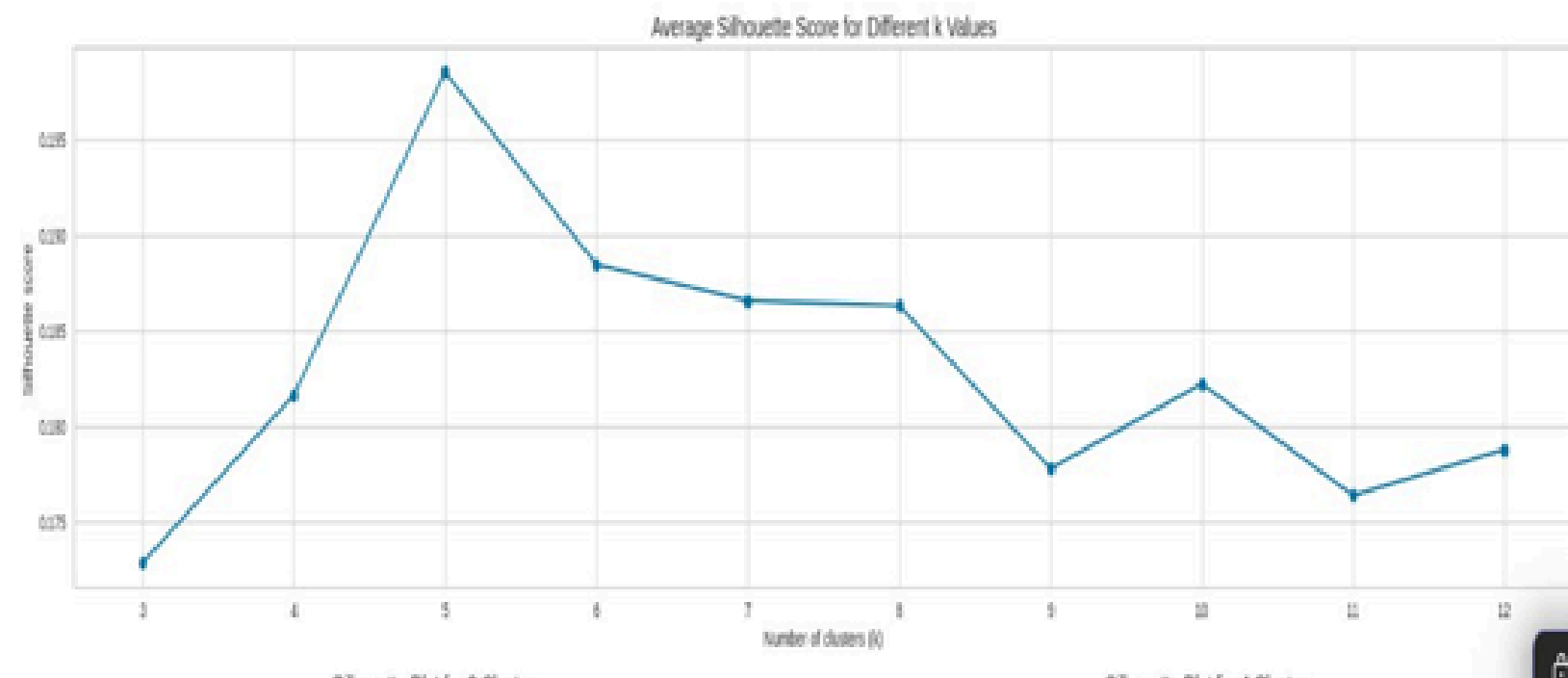
Basket Behaviour

K-Means — Choosing the Optimal k

Elbow Method → k = 4 or 5



Silhouette Score → peak at k = 5



Why k = 4 and not k = 5 ?

Both methods point to k = 5 mathematically — this is sound. Choosing k = 4 is a deliberate business decision.

Interpretability

4 segments map to clear business profiles. A 5th often splits one group into two near-identical sub-groups.

Actionability

Marketing can realistically run 4 strategies. A 5th adds complexity without proportional value.

Cluster balance

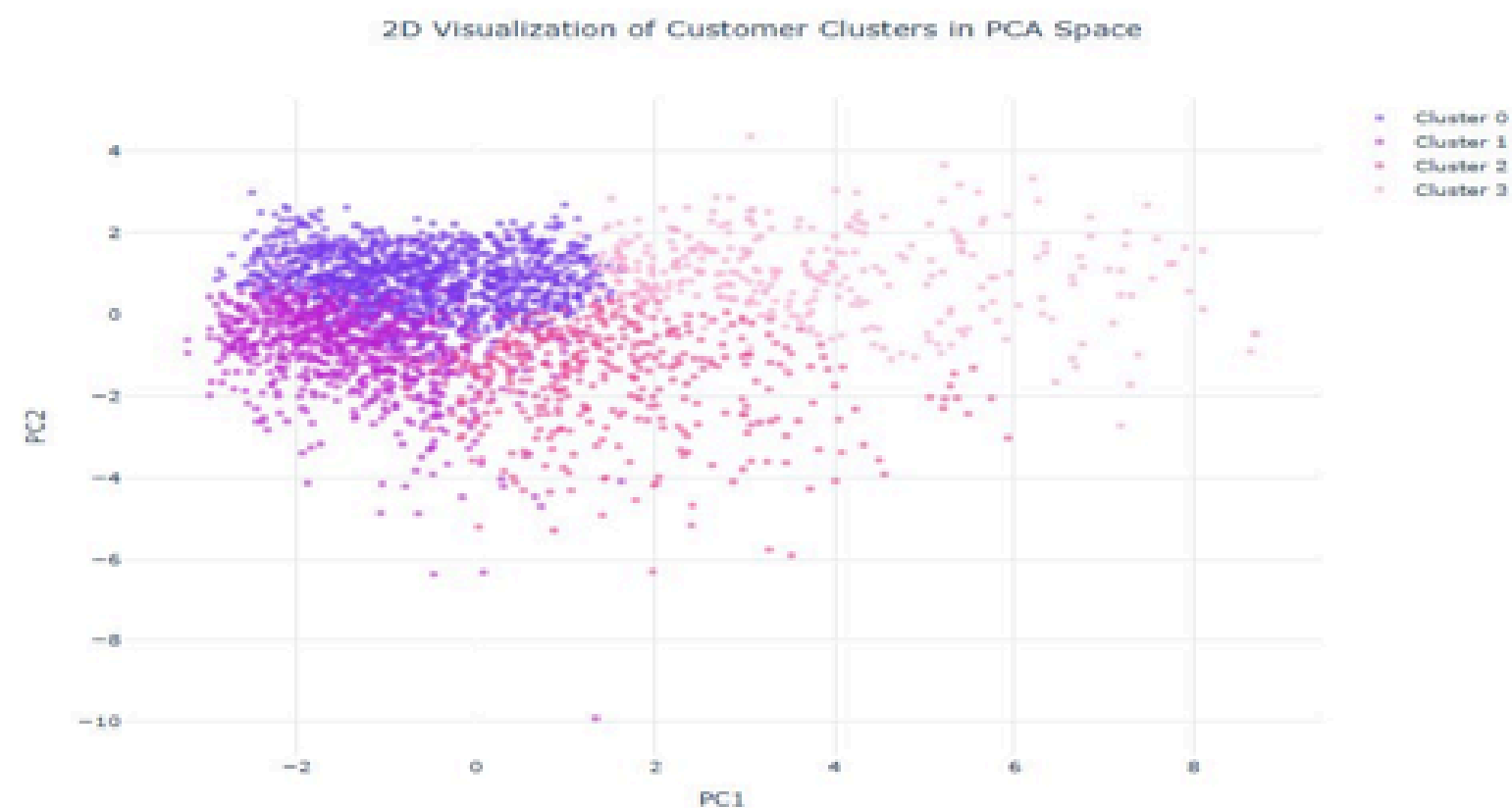
At k = 5, one cluster holds < 8% of customers — too small for a dedicated strategy.

Marginal silhouette gain

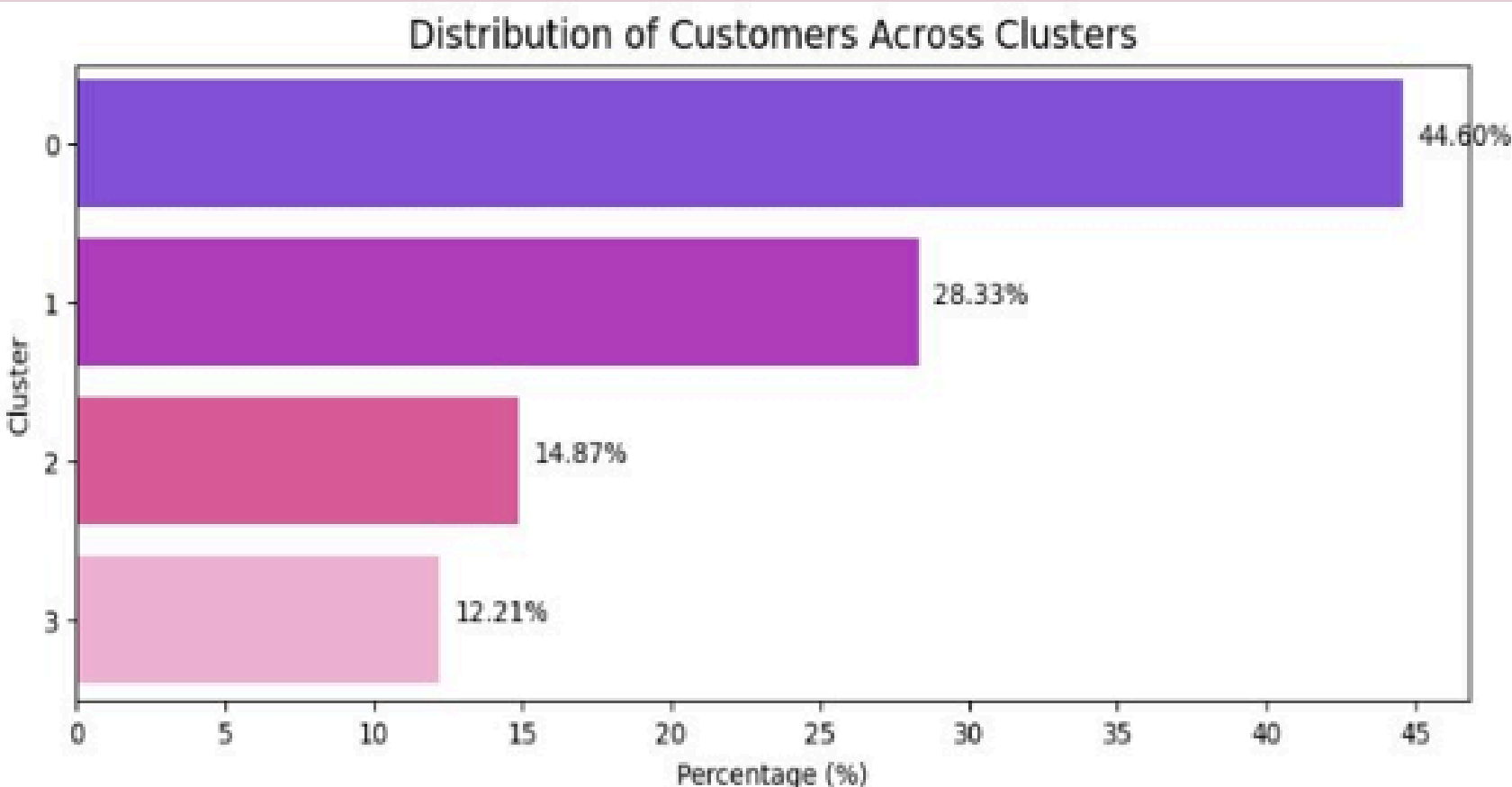
Score difference k=4 → k=5 is < 0.02, not significant enough to override business logic.

k = 5 is mathematically marginal · k = 4 is strategically optimal

2D PCA Cluster Visualization



Customer Distribution Across Clusters



Cluster 0 ~44.6%

Loyal Mid-Value

Recent purchases, moderate frequency, healthy spend. High retention potential.

Strategy

Upsell & cross-sell — nudge toward Champion status.

Cluster 1 ~28.3%

At-Risk / Dormant

Long since last purchase, very few orders, minimal spend. Disengaged.

Strategy

Urgent win-back emails with strong incentives.

Cluster 2 ~14.9%

Champions

Most recent, highest order count, largest spend, widest product variety.

Strategy

Loyalty rewards, VIP programmes, exclusive early access.

Cluster 3 ~12.2%

High-Value Occasional

Infrequent orders but very high transaction value. Likely B2B / bulk buyers.

Strategy

Re-engagement around Q4 peaks. Reduce inter-purchase intervals.

Recommendation System :

Cluster 0 ~44.6%

Loyal Mid-Value

Similar Customer Matching

 Suggest products from similar buyers

↑ Offer better versions of what they buy

 Recommend product bundles

 Move them up to top buyer status

Cluster 1 ~28.3%

At-Risk / Inactive

Win-Back Campaign

 Show their last viewed products

 Send time-limited offers

 Highlight popular items they missed

 Re-engage before full churn

Cluster 2 ~14.9%

Champions

Personalised Discovery

 First access to new arrivals

 Products they haven't tried yet

 Exclusive rewards for top buyers

 Keep them loyal and engaged

Cluster 3 ~12.2%

High-Value Occasional

Basket Pattern Analysis

 Bulk packs of what they buy

 Reminders before Q4 season

 Products often bought together

 Get them to buy more often

Recommendation System :

Regular Browsers

- Macramé Plant Hanger
- Vintage-Style Coasters (Set of 4)
- Woven Seagrass Storage Basket
- String Lights w/ Paper Lanterns
- Reversible Cotton Tote Bag

Home decor · Functional · Eco-friendly

Lapsed Low-Spenders

- Vintage-Style Tea Infuser
- Macramé Plant Hanger
- Personalized Apron
- Floral Patterned Table Runner
- DIY Candle Making Kit

Baking · Crafts · Re-engagement

Power Shoppers

- Vintage-Style Tea Set
- Set of 6 Glass Jars with Lids
- Floral Garland
- Cake Decorating Kit
- Personalized Apron

Retro · Baking · Entertaining

High-Value Inactives

- Vintage-Style Tea Set
- Colorful String Lights
- Flower-Shaped Cookie Jar
- Geometric Cushion Cover
- Hanging Floral Garland

Retro · Decor · Win-back



Thank You

for your attention

Questions & Discussion Welcome